

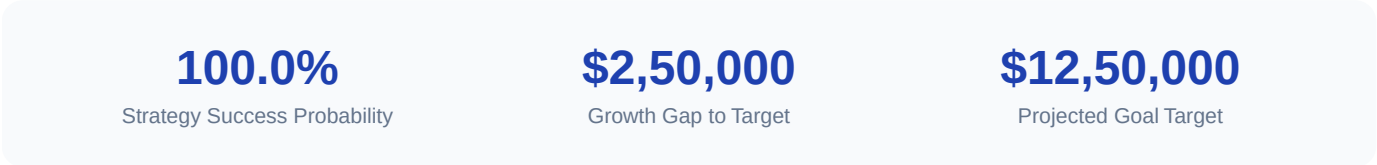
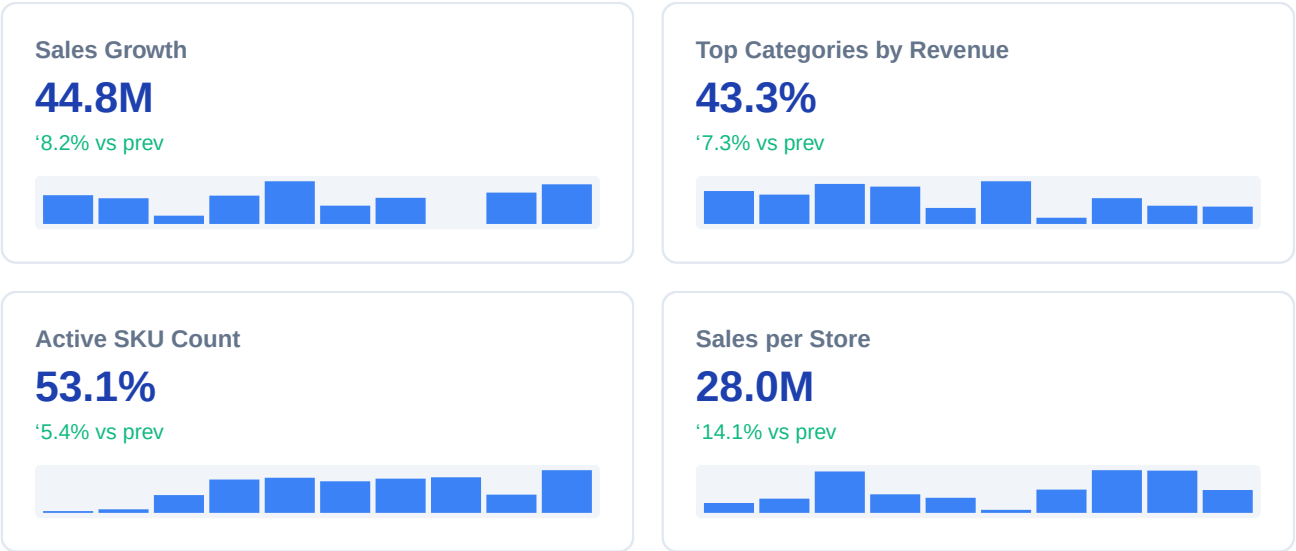
VistaraBI Strategic Intelligence Report

BUSINESS DOMAIN: RETAIL

EXECUTIVE SUMMARY

Comprehensive strategic intelligence report for archive (7). Analyzed 6 KPIs across 5 sources.

OPERATIONAL INTELLIGENCE DASHBOARD (MODULE 5)



STRATEGIC ROADMAP (MODULE 7)

- Omnichannel Expansion Starts**
Strategy Milestone for Day 5: Projected success uplift initiated.
- Omnichannel Expansion Ramp Complete**
Strategy Milestone for Day 15: Projected success uplift initiated.
- Dynamic Pricing Starts**
Strategy Milestone for Day 15: Projected success uplift initiated.
- Dynamic Pricing Ramp Complete**
Strategy Milestone for Day 25: Projected success uplift initiated.
- Loyalty Program Launch Starts**
Strategy Milestone for Day 25: Projected success uplift initiated.
- Loyalty Program Launch Ramp Complete**
Strategy Milestone for Day 35: Projected success uplift initiated.
- Inventory Optimization Starts**
Strategy Milestone for Day 35: Projected success uplift initiated.
- Inventory Optimization Ramp Complete**
Strategy Milestone for Day 45: Projected success uplift initiated.

- **Flash Sales Event Starts**
Strategy Milestone for Day 45: Projected success uplift initiated.
 - **Flash Sales Event Ramp Complete**
Strategy Milestone for Day 50: Projected success uplift initiated.
 - **Targeted Social Ads Starts**
Strategy Milestone for Day 55: Projected success uplift initiated.
 - **Targeted Social Ads Ramp Complete**
Strategy Milestone for Day 65: Projected success uplift initiated.
 - **Store Layout Update Starts**
Strategy Milestone for Day 65: Projected success uplift initiated.
 - **Store Layout Update Ramp Complete**
Strategy Milestone for Day 75: Projected success uplift initiated.
 - **Vendor Renegotiation Starts**
Strategy Milestone for Day 75: Projected success uplift initiated.
 - **Cross-selling Promos Starts**
Strategy Milestone for Day 80: Projected success uplift initiated.
 - **Vendor Renegotiation Ramp Complete**
Strategy Milestone for Day 85: Projected success uplift initiated.
 - **Referral Program Starts**
Strategy Milestone for Day 85: Projected success uplift initiated.
 - **Goal Target: 1250000**
Strategy Milestone for Day 90: Projected success uplift initiated.
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ANALYTIC DIALOGUE (MODULE 6 Q&A)

Q1: Can we immediately compute Year-over-Year Sales Growth using the current `train` table structure?

A1: Structurally, yes, provided `Date` is cast to `TIMESTAMP` and `Weekly\_Sales` is `DECIMAL`. However, the schema lists `train` with generic features; we must verify `Weekly\_Sales` exists as a distinct column and not embedded in the '+12 more' to ensure type safety.

Q2: The `stores` table schema lists `Temperature` and `Fuel\_Price`; is this normalized correctly for store metadata?

A2: No, this is a structural anti-pattern. `stores` should be a dimension table containing static attributes (Type, Size), while `Temperature` and `Fuel\_Price` are time-series facts belonging in the `features` or `train` fact table to maintain 3rd Normal Form.

Q3: To analyze seasonality, do we have a dedicated Date Dimension table linked to the `Date` column in `train`?

A3: Currently, no. The schema relies on raw `Date` strings or objects. For robust seasonality ETL, we need to ingest a Calendar Dimension table to join on `Date` for extracting Quarter, Month, and Holiday flags structurally.

Q4: Can we join `train` and `features` tables to enrich sales data with fuel prices?

A4: Yes, the composite key (`Store`, `Date`) appears consistent across both tables. However, we must enforce referential integrity constraints during ETL to prevent orphaned records if dates misalign between the two sources.

Q5: Is the data structure sufficient to calculate Gross Margin or Profitability KPIs?

A5: No. The schema lacks `Cost`, `COGS`, or `Profit` columns. Structurally, we cannot derive these KPIs without ingesting a separate Product Cost table or adding these fields to the fact table during ingestion.

Q6: What is the expected data type for the `Markdown` columns during transformation?

A6: They should be standardized to `DECIMAL(10,2)` with NULL handling logic. The schema indicates multiple Markdown columns; an ETL pivot transformation is recommended to normalize these into a single `Promotion\_Type` column for better aggregation.

Q7: How should we handle the `Store` ID column for joining with external regional data?

A7: `Store` should be enforced as an `INTEGER` primary key. Currently, it acts as a foreign key, but without a dedicated Store Dimension table containing Region or Manager info, structural joins to external regional data will fail.

Q8: For the Sales per Store KPI, is the current granularity of `Weekly\_Sales` appropriate?

A8: Yes, the grain is Weekly per Store. To calculate daily Sales per Store, an ETL disaggregation logic would be required, which introduces estimation errors; structurally, we should ingest daily transaction data if daily KPIs are needed.

Q9: Are there any partitioning strategies recommended for the `train` table given the `Date` column?

A9: Absolutely. The `Date` column should be used for range partitioning in the data warehouse. This optimizes query performance for time-bound analysis like Sales Growth without scanning the entire historical dataset.

Q10: What is the final structural recommendation to enable the full KPI summary currently showing N/A?

A10: Implement a Star Schema: create `Fact\_Sales` (from `train`), `Dim\_Store` (cleaned `stores`), and `Dim\_Date`. Add `Cost` columns to `Fact\_Sales` and enforce strict data typing on `Weekly\_Sales` to resolve the N/A KPI status.

PREDICTIVE FORECAST VISUALIZATION (MODULE 8)

KPI: Sales per Store | Trend: Bullish (25% Projected Growth)
Confidence Score: 40% Reliability

